

ME1103

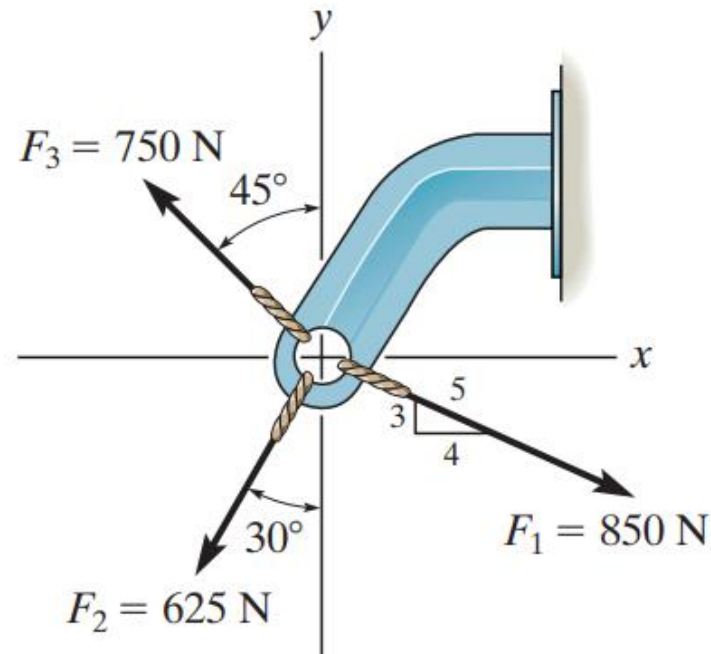
Principles of Mechanics and Materials

PROBLEM SET 1

Question 1

Determine the magnitude of the resultant force and its direction measured counterclockwise from the positive x axis.

Answer: $F_R = 546 \text{ N} @ 253^\circ$

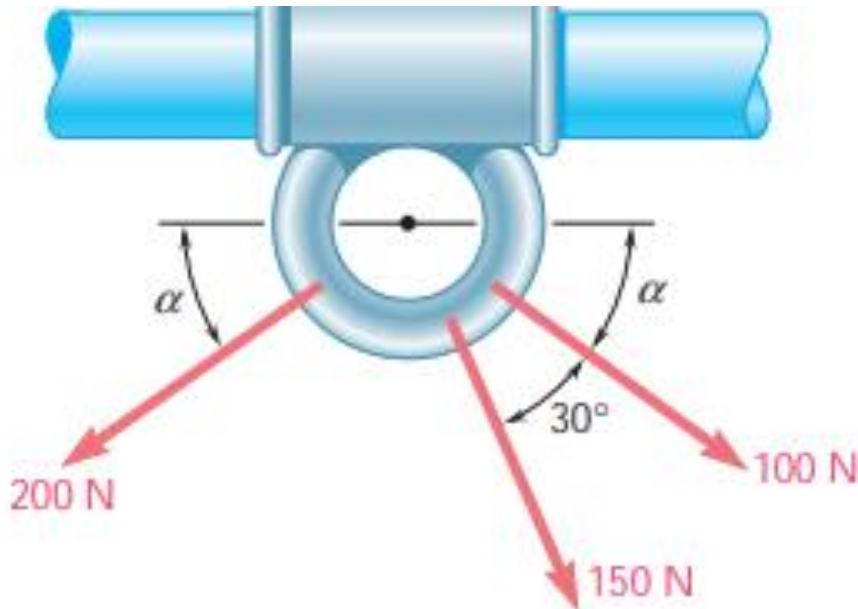


Question 2

Determine (a) the required value of α if the resultant of the three forces shown is to be vertical, (b) the corresponding magnitude of the resultant.

Answer: (a) $\alpha = 21.7^\circ$, and

(b) $R = 229 \text{ N}$



SUM/DIFFERENCES IDENTITIES

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

SUM TO PRODUCT IDENTITIES

$$\sin \alpha + \sin \beta = 2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$$

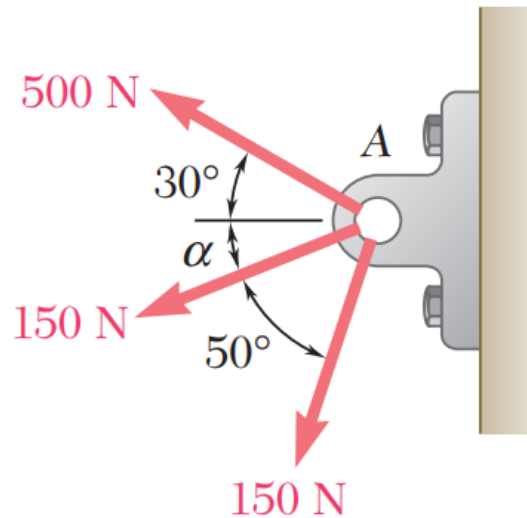
$$\sin \alpha - \sin \beta = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$$

$$\cos \alpha + \cos \beta = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$$

$$\cos \alpha - \cos \beta = -2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$$

Question 3

Three forces are applied on a bracket as shown. The directions of the two 150 N forces may vary, but the angle between these forces is always kept at 50° . Determine the smallest positive α for which the magnitude of the resultant force acting at A is equal to 600 N.



Hints:

What is the resultant force of the two 150 N forces w.r.t. the 500 N force?

Sine and cosine rules can be useful here.

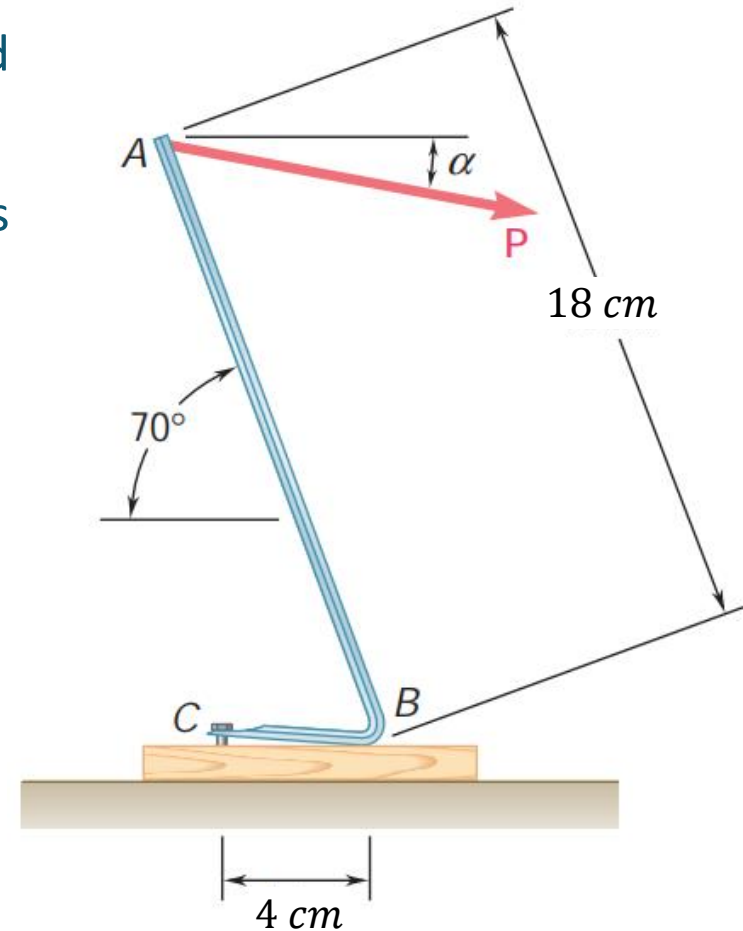
Answer: $\alpha = 27.4^\circ$

Question 4

It is known that a vertical force of 200 N is required to remove the nail at C from the board. Determine:

- the magnitude of the force P that is required to perform the task if $\alpha = 10^\circ$, and
- the smallest force P that can accomplish this task.

*Answer: (a) $P = 51.3\text{ N}$,
 and (b) $P = 44.4\text{ N} \pm 20^\circ$*



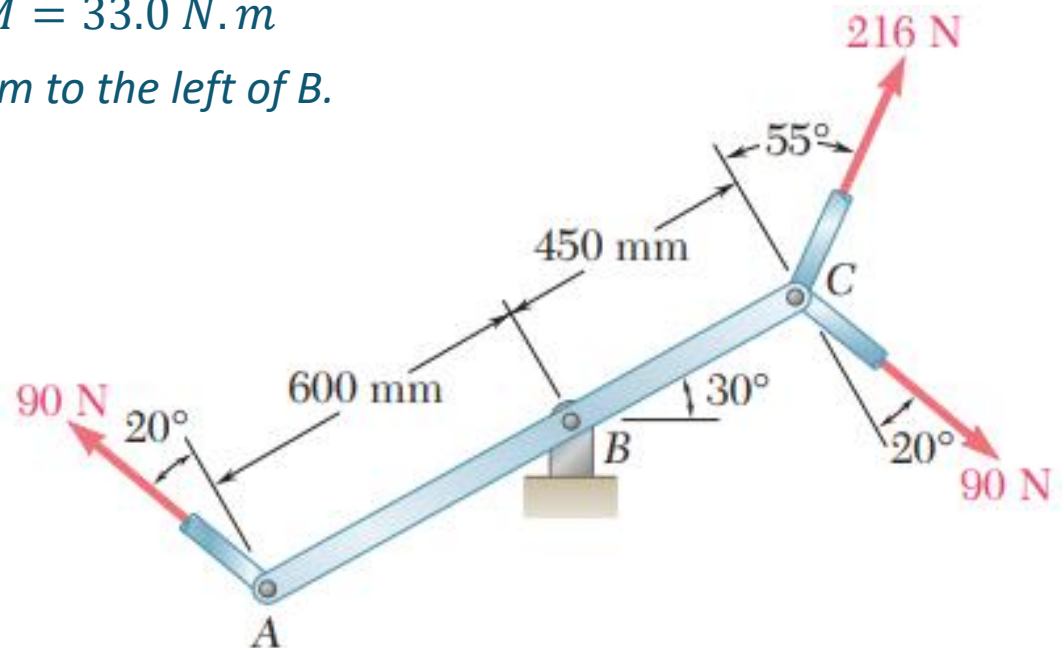
Question 5

Three control rods attached to a lever ABC exert on it the forces shown.

- Replace the three forces with an equivalent force-couple pair at B , and
- Determine the single force that is equivalent to the force-couple system obtained in (a), and specify its point of application on the lever.

Answer: (a) $F = 216\text{ N} \angle 65^\circ$; $M = 33.0\text{ N.m}$

(b) Force shifted by 267 mm to the left of B .



Question 6

Determine a single force that can be applied on the rod AB , such that it will give rise to zero resultant at the fixed end at A .

Answer: $F_R = 542 \text{ N}$ $\angle -10.6^\circ$ from horizontal, and applied at 0.827 m from A .

